



A pneumatic net gun used to shoot a net at an intruding drone (Source: <https://newatlas.com>)

point location, or a predefined set of boundaries. In geofence neutralisation the target drones are prevented from approaching the pre-defined point or entering the defined boundaries, thus blocking them from trespassing.

Geofencing could be dynamic or static. In dynamic geofencing the information regarding restricted area is continuously propagated. Static geofencing uses a flight permission information repository, which when accessed by the intruding drone denies it access beyond the intended point.

Most commercial drones use internal auto-landing modules for safety. Geofencing is unable to disable these automatic landing systems.

Killer drones. Killer drones are employed to track and destroy the invading drones by physically striking them down. This technique requires reactive and real-time decision making, accurate target flying path estimation, and outstanding physical durability and mobility. Swarming killer drones with distributed intelligence and precise tracking systems are promising solution for drone fleet multi-faceted attacks.

High-power microwave pulses. High-power microwave (HPM) devices generate electromagnetic pulses (EMPs) capable of disrupting electronic devices. The EMPs interfere with the radio links and disrupt or even destroy the electronic circuitry by inducing damaging voltages and currents. As a result, the targeted

drone falls uncontrolled instantly.

HPM devices use an antenna to focus the EMPs in the required direction to reduce the potential collateral damage. Within the range, EMPs provide effective non-destructive neutralisation, but have high cost, and create risk of unintentionally disrupting communications or destroying other electronic devices in the area.

High-energy lasers. High-energy lasers use high-powered extremely focused laser beams to destroy the structure and/or the electronics of the targeted drone. They are high-cost, carry risk of collateral damage and are bulky in size.

Nets and net guns. Firing a net at a targeted drone, or bringing a net into contact with it immobilises it by prohibiting the rotor blades from moving. Mainly, they are of three types:

1. *Net cannons fired from the ground.* These could be hand-held,



A counter unmanned aerial system (CUAS) (Source: www.armyrecognition.com)

shoulder-launched, or turret-mounted. They have limited effective range, anywhere from 20m to 300m, and can be used with or without a parachute for controlled descent of the captured drone.

2. *Net cannons fired from another drone.* These overcome the limited range of those fired from the ground and are normally used with a parachute for controlled descent of the captured drone.

3. *Hanging net deployed from a 'net drone.'* The targeted drone is captured by manoeuvring a net-carrying drone towards it. The 'net drone' is normally capable of carrying the captured drone to a safe zone. However, if it is too heavy, the captured drone is released with a parachute for controlled descent, or it is allowed to crash. Physically capturing the drone facilitates a forensic examination and prosecution.

The ground-launched net cannons are semi-automatic with high accuracy and have a short range. The drone-deployed nets provide long range and low risk of collateral damage. Drone-deployed nets are imprecise and have long reload time.

Birds of prey. This technique takes advantage of the natural hunting instinct of the eagles, used by man for hunting for thousands of years. It is possible to train eagles to capture drones. This low-tech solution requires a lot of manpower and time for training and maintenance of these birds of prey (at least one year per bird).

Interception of the drone by these birds of prey can be quick and accurate with low risk of collateral damage. They are difficult to scale due to limited number of birds available and could be an air hazard at airports.

Integrated counter-drone systems. Integrated counter-drone systems are a match-and-mix of the technologies, depending on the specific